



Lesson 5.2 — Square Root Functions

Big idea: $f(x) = \sqrt{x}$ is the parent square-root function. Every transformation (shift, stretch, reflect) changes the starting point and direction of its graph in predictable ways.

Quick review

Parent: $f(x) = \sqrt{x}$. Domain $[0, \infty)$, range $[0, \infty)$, starts at $(0, 0)$.

Transformations: $g(x) = a\sqrt{b(x - h)} + k$ starts at (h, k) and is stretched/reflected by a and b .

Shape: always concave down (for positive a and the standard orientation), increasing.

To find domain: set the radicand ≥ 0 and solve.

Worked example

Worked example. Graph $f(x) = 2\sqrt{x + 3} - 1$. State domain and starting point.

Inside: $x + 3 = x - (-3) \rightarrow$ shift left 3. Outside: vertical stretch by 2, shift down 1. Starts at $(-3, -1)$. Domain: $x \geq -3$. Range: $y \geq -1$.

Warm-ups

1. State the domain of $f(x) = \sqrt{x - 4}$.
2. What is the starting point of $f(x) = \sqrt{x + 1} + 5$?
3. What is the range of $f(x) = \sqrt{x} - 7$?

Practice

1. State the domain and range of $f(x) = \sqrt{x - 2} + 3$.
2. State the domain and range of $f(x) = -\sqrt{x + 1} + 4$.
3. Sketch (describe) $f(x) = 3\sqrt{x - 1}$.

Challenge

1. Find the inverse of $f(x) = \sqrt{x + 5}$ and state its domain.
2. A water tank empties so that the depth (in inches) at time t (minutes) is given by $d(t) = \sqrt{36 - 4t}$. When does the tank become empty? Sketch (describe) the graph.



Answer key — Lesson 5.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x \geq 4$.
2. $(-1, 5)$.
3. $y \geq -7$.

Practice

1. Domain: $x \geq 2$. Range: $y \geq 3$.
2. Domain: $x \geq -1$. Range: $y \leq 4$ (reflection makes it decrease from 4 downward).
3. Shift right 1, vertical stretch by 3. Starts at $(1, 0)$, grows like $3\sqrt{x}$.

Challenge

1. $y = \sqrt{x + 5} \Rightarrow$ swap $x = \sqrt{y + 5} \Rightarrow y = x^2 - 5$, but with $x \geq 0$ (since the original output is non-negative).
 $f^{-1}(x) = x^2 - 5$ with domain $x \geq 0$.
2. Empty when $36 - 4t = 0 \Rightarrow t = 9$ min. Graph starts at $(0, 6)$, decreases ($\sqrt{\text{of a decreasing radicand}}$), reaches 0 at $t = 9$. It's a half-parabola lying on its side, reflected.